Tuning report: tropical / Quickselect

2026-10-22T19:17:08

1 Decision

Approving freezes `fishr.lambda` = 10 into every cell of the grid (frozen hash `089aca544ed4`). The experiment YAML has `fishr.lambda` = 1000 for this campaign. `frozen.yaml` is currently unapproved with {'fishr.lambda': 10.0}.

Selection rule. The candidate with the lowest mean `id_val_loss` wins, averaged over 6 cell(s) and 2 tuning seed(s) each (global seed 1000, disjoint from the grid's 0), across 4 candidates. A candidate with a missing or diverged run is not eligible.

Verdict: REVIEW

- **review**: fishr.lambda=10 is the smallest value searched: the optimum may lie beyond it, so widen the space
- **review**: 12 of 12 winner runs reached their best id_val_loss at the last logged epoch: the search budget may be too short to rank candidates
- **review**: 12 of 12 winner runs picked their last epoch (best_epoch = 19): still improving when the search budget ran out, so candidates were ranked before converging
- **info**: the search ran with optim.epochs=20; the grid runs with optim.epochs=100
- **info**: devices: cuda:0

block: do not approve; fix the search and re-run it. **review**: read the sections below before approving. **info**: context only.

2 Search setup

- `fishr.lambda` ∈ {10, 100, 1000, 1e+04}
- Cells trained (`tune.only` {'objective': 'fishr'}):
  - `model_type=vanilla head=seq2seq objective=fishr`
  - `model_type=vanilla head=pointer objective=fishr`
  - `model_type=adaptive head=seq2seq objective=fishr`
  - `model_type=adaptive head=pointer objective=fishr`
  - `model_type=tropical head=seq2seq objective=fishr`
  - `model_type=tropical head=pointer objective=fishr`
- Search-only overrides (never frozen): {'optim': {'epochs': 20}}. The chosen value was ranked under this budget, not the grid's.

3 Candidates

Table 1 ranks the candidates on `id_val_loss` (lower is better); Figure 1 plots them. Candidates share seeds, so Table 2 compares each with the winner run by run (same cell, same seed).

Table 1: Every candidate: rank, runs stored/expected, diverged, mean with 95% bootstrap CI, SD and range over all its runs.

#	`fishr.lambda`	runs	div.	mean	95% CI	SD	range	
1	10	12/12	0	0.1495	[0.0799, 0.224]	0.135	[0.0312, 0.364]	**selected**
2	100	12/12	0	0.1534	[0.0824, 0.228]	0.138	[0.0311, 0.364]	
3	1000	12/12	0	0.1809	[0.108, 0.262]	0.144	[0.0321, 0.393]	
4	1e+04	12/12	0	0.2051	[0.132, 0.275]	0.136	[0.0405, 0.401]	

Table 2: Each candidate minus the winner on the runs they share, signed so a positive gap is worse than the winner. Worse/better/equal count the runs; p is a two-sided sign test.

candidate	n	gap	95% CI	worse	better	equal	p
`fishr.lambda=100`	12	0.003921	$[9.12 \times 10^{-4}, 0.00801]$	10	2	0	0.0386
`fishr.lambda=1000`	12	0.03142	[0.0139, 0.0517]	12	0	0	0.0005
`fishr.lambda=10000`	12	0.05558	[0.0281, 0.0839]	11	1	0	0.0063

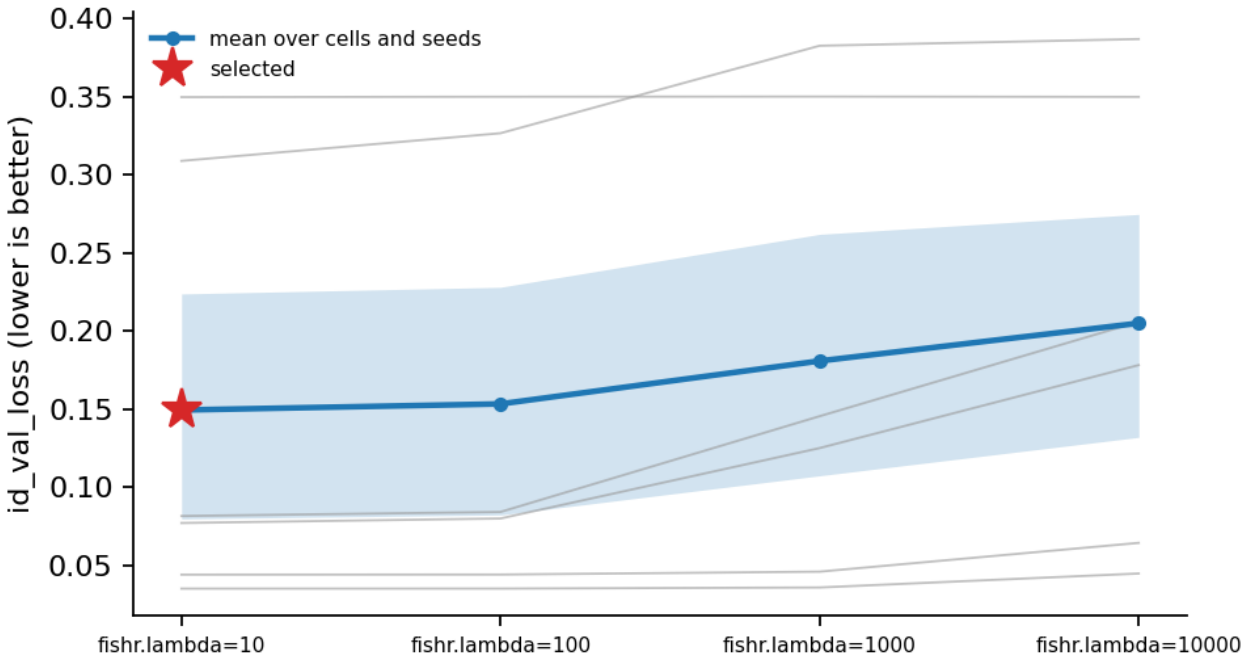


Figure 1: `id_val_loss` per candidate: the mean over every run with a 95% bootstrap CI (blue), each cell's seed-mean (grey), the selected candidate (red star).

4 Per cell

One value is frozen for every cell, chosen by the mean over cells. Figure 2 shows whether the winner is best in each cell or is carried by a few of them.

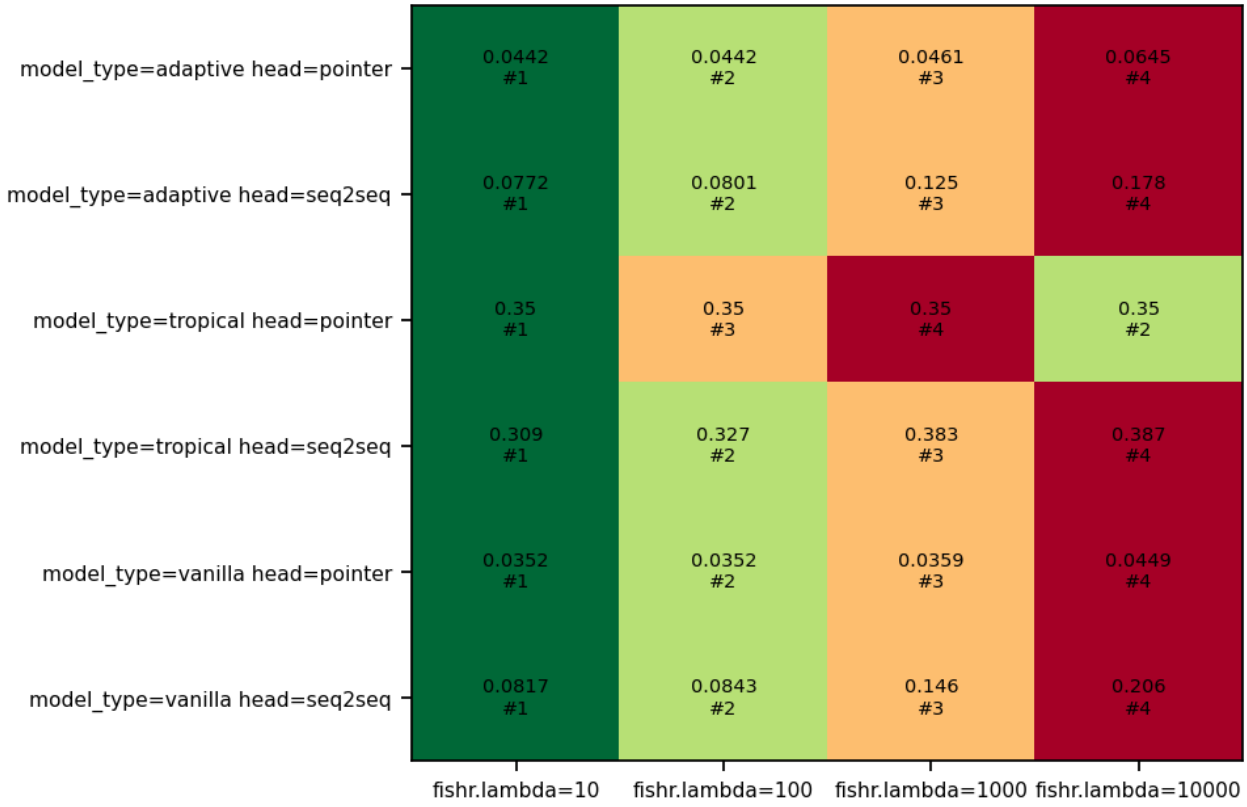


Figure 2: Each cell's seed-mean `id_val_loss` per candidate, with its rank within the cell (#1 is best; green is better).

Table 3: The best candidate in each cell.

cell	best
`model_type=adaptive head=pointer`	fishr.lambda=10
`model_type=adaptive head=seq2seq`	fishr.lambda=10
`model_type=tropical head=pointer`	fishr.lambda=10
`model_type=tropical head=seq2seq`	fishr.lambda=10
`model_type=vanilla head=pointer`	fishr.lambda=10
`model_type=vanilla head=seq2seq`	fishr.lambda=10

5 Other training-distribution metrics

Not used for selection; shown so a winner that trades them away is visible.

Table 4: Mean of every other in-distribution metric per candidate.

candidate	`id_f1`	`id_exact`	`fishr_penalty_final`	`best_epoch`
fishr.lambda=10	0.887	0.772	6.65×10^{-4}	19

Table 4: Mean of every other in-distribution metric per candidate.

candidate	id_f1	id_exact	fishr_penalty_final	best_epoch
fishr.lambda=100	0.882	0.768	1.95×10^{-4}	19
fishr.lambda=1000	0.866	0.721	8.25×10^{-5}	17.9
fishr.lambda=10000	0.839	0.683	5.38×10^{-5}	12.8

6 Training curves

The mean over every cell and seed of each candidate; the selected candidate in bold red.

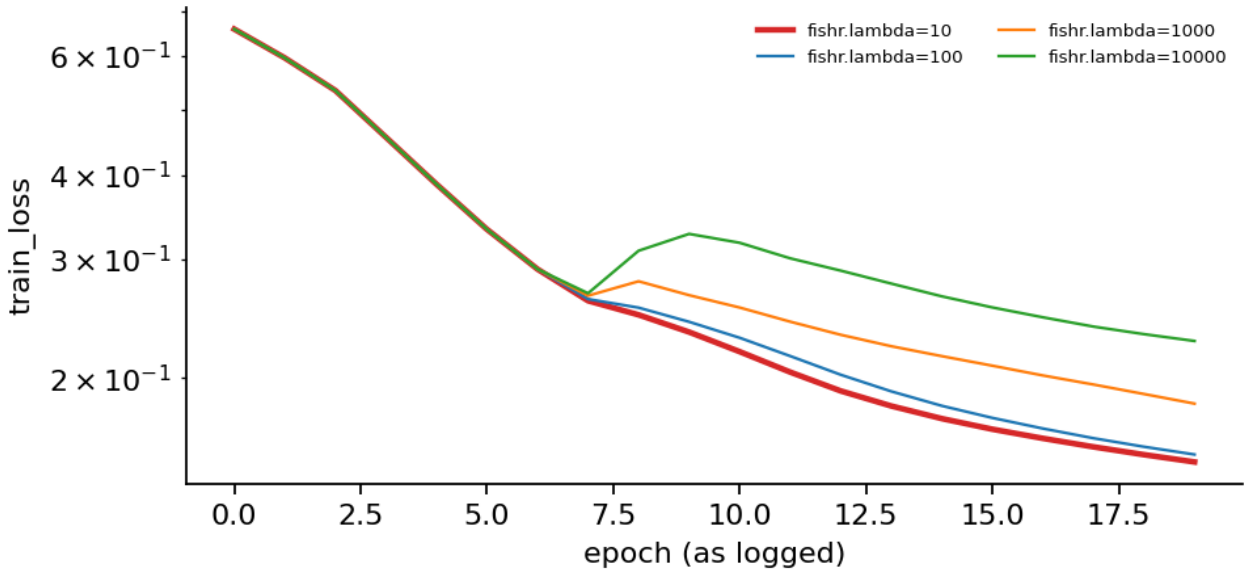


Figure 3: `train_loss` per candidate.

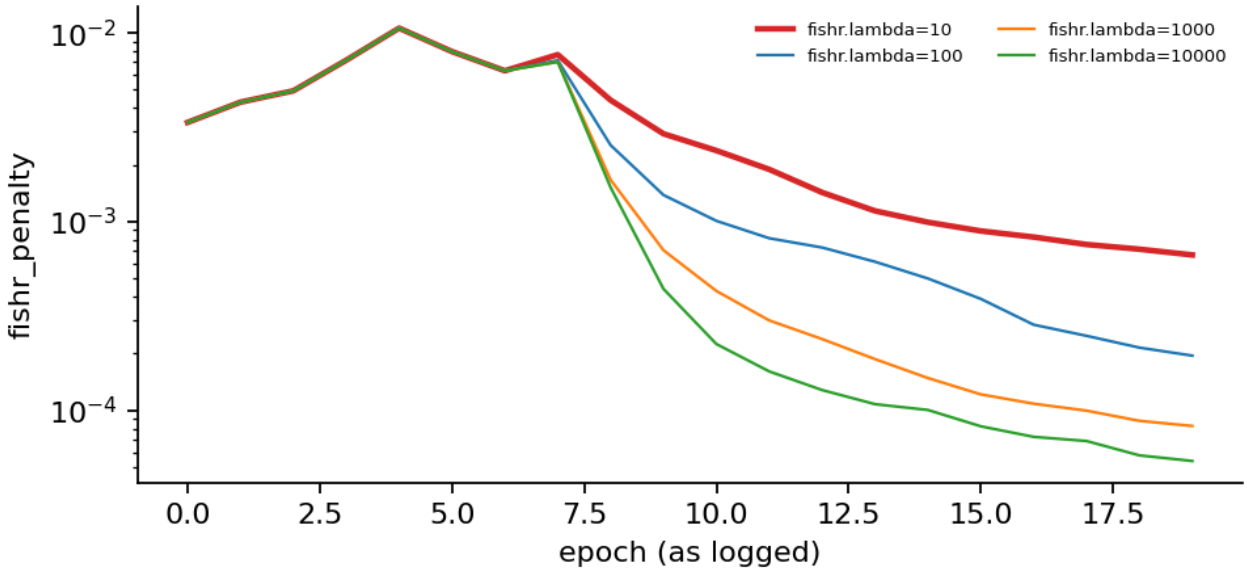


Figure 4: `fishr_penalty` per candidate.

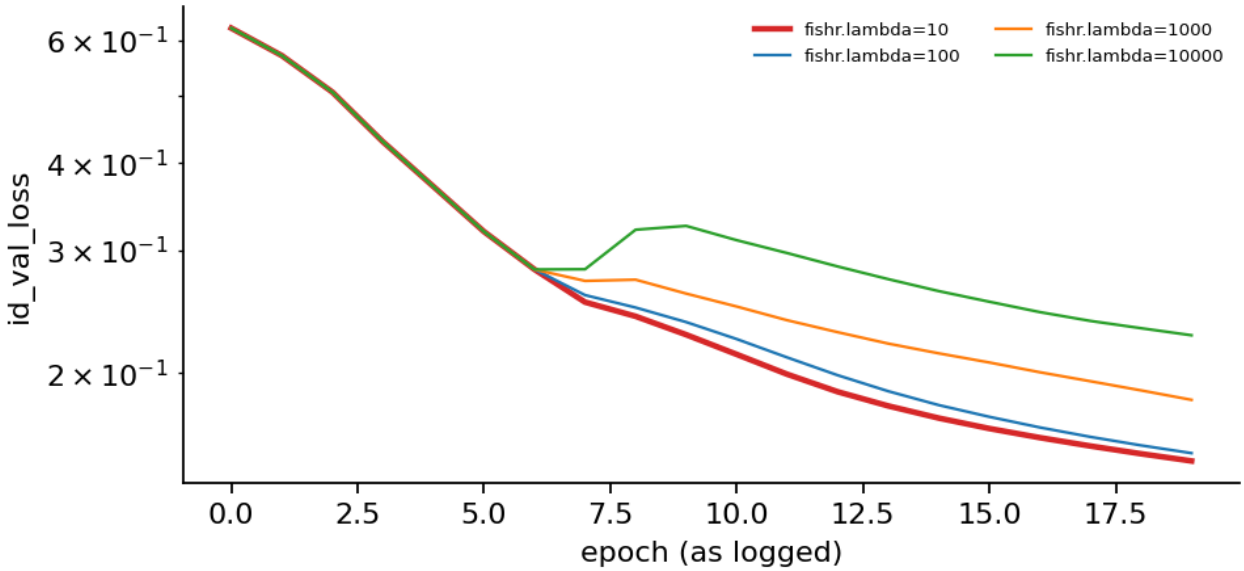


Figure 5: `id_val_loss` per candidate.

7 Provenance

Table 5: Each candidate's runs: base hash, runs, total wall time, commits (+ marks a dirty tree), torch.

candidate	hash	runs	wall (s)	commit	torch
fishr.lambda=10	`333e58e36b41`	12	2.83e+03	`e5ccce8034`	2.8.0+cu128
fishr.lambda=100	`e57d0bde68db`	12	2.84e+03	`e5ccce8034`	2.8.0+cu128
fishr.lambda=1000	`231db5f871fc`	12	3.61e+03	`e5ccce8034`	2.8.0+cu128
fishr.lambda=10000	`6d2b2749853a`	12	3.6e+03	`e5ccce8034`	2.8.0+cu128

Store: `campaigns/tropical/Quickselect/tune/results`.
Rebuild: `expkit tune-report --config configs/tropical.yaml --profile Quickselect`
Approve: `expkit approve --config configs/tropical.yaml --profile Quickselect`

The base the grid would run with.

```
task:
  name: QuickselectDataset
  value_range:
  - 1
  - 10
  target_range: null
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - 11
  - 21
  ood_noise_range:
  - 1
  - 5
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  lambda: 10.0
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

A Sealed: OOD and test metrics

Read this only after deciding. These were recorded by the tuning runs but never consulted by the selection. Choosing a value because it looks better here would tune on the test distribution.

Table 6: Mean of every OOD or test metric per candidate.

candidate	ood_len_f1	ood_len_exact	ood_val_f1	ood_val_exact	ood_noise_f1	ood_noise_exact	ood_len_f1_L12	ood_len_exact_L12	ood_len_f1_L16	ood_len_exact_L16	ood_len_f1_L64	ood_len_exact_L64
fishr.lambda=10	0.599	0.205	0.881	0.758	0.45	0.179	0.731	0.299	0.692	0.213	0.53	0.103
fishr.lambda=100	0.553	0.202	0.876	0.753	0.443	0.177	0.722	0.299	0.682	0.224	0.446	0.0825
fishr.lambda=1000	0.6	0.149	0.859	0.704	0.444	0.183	0.72	0.25	0.688	0.172	0.543	0.0252
fishr.lambda=10000	0.632	0.143	0.832	0.667	0.437	0.193	0.705	0.23	0.68	0.154	0.599	0.0443